

Dennis' Stock & Options Scanner Playbook

Published specification for five active scheduled scanners, one paused scheduled scanner, and one on-demand scanner

PUT SPREAD | PUT SKEW | WEAK CLOSE | 5-DAY WEAK CLOSE | FRIDAY TO MONDAY | 90-DAY
DOWNSIDE | 3-5 DAY SWING

Published	September 16, 2026
Timezone	America/Los_Angeles (Pacific Time)
Primary market data	Massive
Purpose	Systematic screening and candidate ranking; not trade instructions

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Important: Scanner outputs are statistical screens. They do not guarantee a profitable trade, eliminate gap risk, or replace live order-entry verification.

1. Executive Summary

This playbook consolidates the exact operating logic of Dennis' five active scheduled scanners, the paused 90-Day / 15% Downside Safety Scanner, and the on-demand 3-5 Day Swing Scanner into one publishable reference. The scanners now cover conservative options-premium setups, short-horizon rebound patterns, and momentum/catalyst swing opportunities. Massive remains the primary quantitative market-data source.

Scanner	Runs (PT)	Instrument	Primary Setup	Typical Hold / DTE	Full-Pass Focus
Put Spread	Mon-Fri, 8:30 AM	Put credit spread	Down-day stock + deep OTM spread	30-120 DTE	Low net delta, downside buffer, earnings clearance, credit/RoR
Put Skew	Mon-Fri, 8:30 AM	Put credit spread	Rich downside put skew vs ATM IV/RV	30-120 DTE	Skew premium, IV/RV, low net delta, liquidity
Weak Close	Mon-Fri, 12:40 PM	Stock	Weak close → next-open rebound	Overnight	>=60% next-open win rate + positive avg/median
5-Day Weak Close	Mon-Fri, 12:40 PM	Stock	Weak close → multi-day rebound	Up to 5 trading days	>=65% 5-day win rate + >=80% +1% hit rate
Friday	Fri, 12:40 PM	Stock	Weak Friday → Monday rebound	Friday close to Monday	>=60% Monday open/close wins + >=70% +1% hit rate
3-5 Day Swing	On demand	Stock	Momentum + breakout + catalyst confirmation	3-5 trading days	Multi-factor score + persistence + event-risk control
90-Day Downside	M-F 8:30 AM (paused)	Stock or ETF; optional put spread	Avoid >15% path drawdown over ~90 days	63 sessions / 30-120 DTE	>=95% historical survival plus robustness checks

How to use the scanners together

- 8:30 AM PT: Put Spread and Put Skew scan the options market for conservative premium-selling candidates.
- 12:40 PM PT: Weak Close and 5-Day Weak Close evaluate late-day equity weakness for overnight or multi-day rebound setups.
- Friday 12:40 PM PT: Friday scanner specializes in the weekend-to-Monday effect and uses a separate historical test.
- Paused scheduled task: 90-Day / 15% Downside Safety Scanner is configured for Monday-Friday at 8:30 AM PT when enabled, can be run manually on demand while paused, and emails every completed report to Dennis Weng at dwengca@gmail.com.

- On demand: 3-5 Day Swing screens liquid U.S. stocks for momentum, breakout quality, volume, historical 3-5 day persistence, options positioning, catalysts, and event risk. It is not a scheduled task.
- Full Pass means the candidate satisfies the scanner's hard rules. Watchlist means the setup is close but misses one or more required conditions or contains an unresolved verification item.

2. Shared Definitions & Risk Controls

Classification

- Full Pass — meets all hard gates and verification requirements for that scanner.
- Watchlist — near miss, recent persistence failure, Tier C delta, unverified earnings timing, or other material caveat.
- Reject — fails a hard rule such as delta, liquidity, earnings buffer, historical win-rate threshold, or required option open interest.

Net put-spread delta

Both options scanners use the net delta of the complete vertical spread rather than the short-leg delta as the primary delta gate.

Net spread delta: $(-1 \times \text{short-put delta}) + \text{long-put delta} = |\text{short delta}| - |\text{long delta}|$ (when both put deltas are negative)

Tier	Absolute Net Delta	Treatment
A — Ideal	≤ 0.03	Full-pass eligible
B — Acceptable	> 0.03 to ≤ 0.05	Full-pass eligible
C — Watchlist	> 0.05 to ≤ 0.08	Watchlist only
Reject	> 0.08	Reject

Earnings discipline

Earnings timing is treated conservatively. If the relevant earnings date or time cannot be verified, a candidate cannot receive Full Pass. Projected dates are labeled as projected and the required buffer is still applied.

Data and execution discipline

- Massive is the primary quantitative market-data source.
- Put Spread may use Alpaca as a secondary source for indicative bid/ask and spread economics.
- Indicative or stale prices must be labeled ESTIMATED/INDICATIVE rather than executable.
- Options candidates must be verified against live broker pricing before any trade decision.

3. Put Spread Scanner

Schedule	Monday-Friday at 8:30 AM PT
Primary data	Massive; Alpaca for secondary indicative pricing
Instrument	Bullish put credit spreads
Target expiration	30-120 DTE
Core objective	Find liquid stocks with conservative downside buffers and low net spread delta while maintaining acceptable credit and return on risk.

Universe & stock gates

- U.S. equities priced from \$3 to \$1,000, prioritizing large-cap and S&P 500 / Nasdaq-100 names when appropriate.
- Stock must currently be down on the day.
- During the prior 30 trading days, at least 5 sessions must have absolute daily returns greater than 2%.
- No single daily decline during that lookback may be -4% or worse.
- Average daily volume must be at least 1.5 million shares.

Spread construction & hard gates

- Short put strike must be no more than 80% of the current stock price.
- Short put strike must also be no more than 120% of the 52-week low.
- Use practical widths: generally \$5 for higher-priced stocks, with \$2.50, \$1, or another sensible width for lower-priced names.
- Near-strike option open interest must be at least 500 contracts; report OI on both legs.
- Net spread delta must be Tier A or Tier B for Full Pass. Tier C is Watchlist only; above 0.08 is rejected.

Earnings filter

- A Full Pass spread must expire at least 7 calendar days before the next confirmed or conservatively projected earnings event.
- If a selected expiration crosses the earnings buffer, search another qualifying expiration within 30-120 DTE.
- If no qualifying pre-earnings expiration exists, exclude the stock from Full Pass.
- If earnings cannot be reliably established, classify the otherwise qualifying setup as Watchlist — EARNINGS DATE UNVERIFIED.

Pricing & return-on-risk

Return on risk: net credit / (spread width - net credit)

- Estimated/live net credit must be at least \$0.25 for Full Pass.
- Estimated/live return on risk must be at least 5%.
- Combined indicative bid-ask width of \$0.08 or less is preferred; sparse or distorted indicative data should be flagged rather than invented.

Broker check: VERIFY IN ROBINHOOD BEFORE TRADE. Require live spread credit $\geq$ \$0.25 and live RoR $\geq$ 5%. If live economics fail, do not rely on the scanner result.

Ranking priorities

Rank by overall quality, emphasizing downside buffer, net-delta tier, earnings-event clearance, Massive OI/liquidity, estimated credit, and estimated RoR.

Required output fields

Ticker/company; spot price and daily change; 52-week low; expiration/DTE; next earnings date and confirmed/projected status; days from expiration to earnings; short and long strikes; short delta; long delta; calculated net spread delta and tier; bid/ask on both legs when available; estimated net credit; OI on both legs; IV if available; max risk; estimated RoR; downside buffer to short strike; 30-day volatility check; reasons it passed; verification caveats.

4. Put Skew Scanner

Schedule	Monday-Friday at 8:30 AM PT
Primary data	Massive
Instrument	Put credit spreads
Target expiration	30-120 DTE
Core objective	Find unusually expensive downside puts relative to ATM IV and recent realized volatility, then monetize that skew with conservative vertical spreads.

Universe & liquidity

- Active U.S.-listed common stocks; prioritize large-cap, S&P 500, and Nasdaq-100 names.
- Generally target market cap $\geq$ \$10 billion when reference data is available.
- Exclude ETFs, leveraged/inverse funds, OTC securities, preferreds, warrants, and non-common shares.
- Average share volume over the prior 30 trading days must be at least 1.5 million.

Spread search

- Short-put delta approximately -0.05 to -0.15.
- Short strike generally no more than 85% of spot.
- Practical spread width generally \$5 for higher-priced stocks and \$2.50 or \$1 for lower-priced stocks.
- Open interest must be at least 500 contracts on both legs for Full Pass.

Skew calculations

Put Skew Premium: short-put IV - ATM IV

IV/RV Ratio: short-put IV / RV30

- Estimate ATM IV from the closest-to-ATM liquid options.
- Realized volatility emphasizes the most recent 30 trading days (RV30).

Delta Tier	Skew Premium Target	IV/RV Target	Treatment
A — Ideal	≥ 8 vol points	≥ 1.25	Full-pass eligible
B — Acceptable	5-8 vol points	≥ 1.15	Full-pass eligible
C — Watchlist	Depends on overall quality	Depends on overall quality	Watchlist only
Reject	Below thresholds with weak overall quality	Weak	Reject

Pricing, earnings & event controls

- Estimated/live net credit $\geq$ \$0.25 and RoR $\geq$ 5% for Full Pass.
- Combined spread bid-ask width $\leq$ \$0.08 is preferred; flag sparse/wide pricing.
- For Full Pass, expiration must be at least 7 calendar days before the next confirmed or conservatively projected earnings event.
- Flag known material or binary events before expiration.
- Earnings date unverified → Watchlist, not Full Pass.

Ranking formula

Component	Approx. Weight
Skew premium	30%
IV/RV premium	20%
Downside distance to short strike	20%
Spread RoR	15%
Liquidity / OI	10%
Net delta	5%

Apply earnings/event vetoes after ranking. Penalize poor liquidity, extremely event-driven IV, or severe underlying downside tails that may explain the skew.

Required output fields

Ticker/company; underlying price; expiration/DTE; short/long strikes; short strike as % of spot; short/long IV; ATM IV; skew premium; RV30; IV/RV; short delta; long delta; net spread delta and tier; OI each leg; bid/ask each leg if available; estimated credit; max risk; RoR; earnings date/status/days to earnings; event-risk notes; reason for ranking; Full Pass / Watchlist / Reject.

Broker check: Always state VERIFY IN ROBINHOOD BEFORE TRADE. This is a statistical screening list, not a guarantee of safety or profit.

5. Weak Close Scanner

Schedule	Monday-Friday at 12:40 PM PT (approximately 3:40 PM ET)
Primary data	Massive
Instrument	Common stock
Typical holding period	Near close to next regular-session open
Core objective	Find liquid stocks with a historically persistent weak-close-to-next-open rebound tendency.

Universe & liquidity

- Active U.S.-listed common stocks with market cap $\geq$ \$10 billion.
- Exclude ETFs, leveraged/inverse products, OTC, preferreds, warrants, and other non-common-stock instruments.
- 30-day average daily dollar volume $\geq$ \$100 million and average share volume $\geq$ 1 million.

Current-day setup

Close location: (current price - day low) / (day high - day low)

- Stock must be trading below the regular-session open.
- Close location must be ≤ 0.30 — i.e., in the bottom 30% of the day's high-low range.
- Report open-to-current return and close-location percentile.

Historical pattern test

- Use approximately the prior 252 trading days.
- Historical weak close = close < open and close location ≤ 0.30 .

Next-open return: next trading day official regular-session open / prior close - 1

- At least 40 historical qualifying weak-close observations are required.
- Full Pass: next-open win rate $\geq 60\%$, average next-open return $> 0\%$, and median next-open return $> 0\%$.

Recent persistence

- Independently evaluate approximately the most recent 60 trading days.
- Require at least 10 recent qualifying observations and next-open win rate $\geq 60\%$.
- Report recent average overnight return.

Earnings, event & tail-risk controls

- Exclude a Full Pass when confirmed/projected earnings occur within the next 7 calendar days, including tonight or tomorrow.
- Unverified earnings date $\rightarrow$ Watchlist.
- Flag known scheduled material/binary events before the next open.
- Penalize patterns dominated by a few large positive gaps; report worst overnight return and preferably the 5th-percentile overnight return.

Required output fields

Ticker/company; market cap; current price; day open/high/low; open-to-current %; close-location %; 1-year sample count; 1-year next-open win rate; average and median next-open return; worst and 5th-percentile overnight return; recent ~60-day sample/win rate/average; next earnings date/status; days to earnings; liquidity metrics; reason it qualified; classification.

6. 5-Day Weak Close Scanner

Schedule	Monday-Friday at 12:40 PM PT (approximately 3:40 PM ET)
Primary data	Massive
Instrument	Common stock
Typical holding period	Near close to within 5 trading days
Core objective	Find weak closes that historically rebound over the following 3-5 trading days.

Universe & current-day setup

- Same universe and liquidity gates as the Weak Close Scanner: market cap $\geq$ \$10B, average dollar volume $\geq$ \$100M, average share volume $\geq$ 1M.
- Current price below the day's open and close location $\leq$ 0.30.
- Report open-to-current % and close-location %.

Historical test

- Use approximately the prior 252 trading days; weak close = close < open and close location $\leq$ 0.30.
- Calculate 3-trading-day and 5-trading-day forward close returns.
- Also calculate whether subsequent daily highs touched +1% and +2% versus the weak-close entry price during the next 5 trading days.
- Require at least 40 historical weak-close observations.

Full-Pass Metric	Threshold
5-day win rate	$\geq$ 65%
Average 5-day return	$>$ 0%
Median 5-day return	$>$ 0%
Probability of touching +1% within 5 days	$\geq$ 80%

Recent persistence

- Use approximately the most recent 60 trading days.
- Require at least 10 recent qualifying observations, recent 5-day win rate $\geq$ 60%, and recent average 5-day return $>$ 0%.
- If long-term Full Pass metrics succeed but recent persistence fails, classify as Watchlist.

Earnings & tail-risk controls

- Exclude confirmed/projected earnings within the next 7 calendar days and account for the entire expected holding window plus a safety buffer.
- If earnings cannot be reliably established, use Watchlist — EARNINGS DATE UNVERIFIED.
- Flag known scheduled material/binary events during the next 5 trading days.
- Report worst 5-day return and preferably 5th-percentile 5-day return. Demote candidates with poor downside tails even when win rate is high.

Required output fields

Ticker/company; market cap; current price; day open/high/low; open-to-current %; close-location %; 1-year sample count; 3-day win rate; 5-day win rate; average and median 5-day return; +1% and +2% hit rates; worst and 5th-percentile 5-day return; recent sample/win/average; earnings date/status/days to earnings; liquidity metrics; reason it qualified; classification.

7. Friday — Friday → Monday Effect Scanner

Schedule	Friday at 12:40 PM PT (approximately 3:40 PM ET)
Primary data	Massive
Instrument	Common stock
Typical holding period	Near Friday close to Monday open/close
Core objective	Find liquid stocks with a statistically persistent Monday rebound after a weak Friday close.

Universe & liquidity

- Active U.S.-listed common stocks with market cap $\geq$ \$10 billion.
- Exclude ETFs, leveraged/inverse products, OTC, preferreds, warrants, and non-common shares.
- Average daily dollar volume $\geq$ \$100 million and average share volume $\geq$ 1 million over the prior 30 trading days.

Current Friday setup

- Evaluate late in the regular Friday session.
- Stock must be trading below Friday's regular-session open.
- Full Pass current setup requires close location ≤ 0.30 . Watchlist may include ≤ 0.40 if historical edge is unusually strong.
- Report Friday open-to-current return and close-location percentile.

Historical Friday-to-Monday test

- Use approximately the prior 2 years where available, emphasizing the most recent 1 year.
- Qualifying historical Friday: close < open and close location ≤ 0.30 .

Monday open return: Monday open / Friday close - 1

Monday close return: Monday close / Friday close - 1

- Track whether Monday intraday high touched +1% and +2% versus Friday close.
- Track Monday worst intraday drawdown versus Friday close.
- Do not mix holiday-shortened weekends where the next session is Tuesday into the primary Monday statistics; label separately if analyzed.
- Require at least 25 qualifying historical Fridays for Full Pass.

Full-Pass Metric	Threshold
Monday-open win rate	$\geq 60\%$
Monday-close win rate	$\geq 60\%$
Average Monday-close return	$> 0\%$
Median Monday-close return	$> 0\%$
Probability of touching +1% Monday	$\geq 70\%$

Recent persistence

- Evaluate the most recent ~12 months and, when sample size permits, ~6 months.
- Require recent Monday-close win rate $\geq 60\%$ and recent average Monday-close return $> 0\%$ for Full Pass.
- If long-term thresholds pass but recent persistence fails, classify as Watchlist.

Earnings, events & tail risk

- Exclude confirmed or projected earnings scheduled after Friday close, over the weekend, or Monday before the intended exit window.
- If timing cannot be verified, classify as Watchlist — EARNINGS DATE/TIME UNVERIFIED.
- Flag known material/binary events over the weekend or Monday morning.
- Report worst Monday-close return and preferably 5th-percentile Monday-close return; penalize severe downside tails or results dominated by a few large positive gaps.

Required output fields

Ticker/company; market cap; Friday current price/open/high/low; open-to-current %; close-location %; historical sample count; Monday-open win rate/average return; Monday-close win rate/average/median return; +1%/+2% Monday hit rates; worst and 5th-percentile Monday return; recent sample/win/average; earnings date/time/status; event-risk notes; liquidity metrics; reason for ranking; classification.

8. 3-5 Day Swing Scanner (On Demand)

Schedule	On demand only - run whenever requested; does not consume a scheduled-task slot.
Primary data	Massive for stock prices, daily bars, RSI/technical data, options snapshots, and news/sentiment where entitled.
Instrument	Common stock
Typical holding period	3-5 trading days
Core objective	Rank liquid stocks with the strongest evidence of near-term upside continuation while avoiding overextended, illiquid, or event-dominated setups.
Default command	Run my 3-5 Day Swing Scanner using Massive.

Universe & liquidity

- Active U.S.-listed common stocks. Exclude ETFs, leveraged/inverse products, OTC securities, preferreds, warrants, units, and other non-common-share instruments.
- Prefer price $\geq \$5$, 30-day average share volume ≥ 1 million, and 30-day average daily dollar volume $\geq \$100$ million. Lower-liquidity names may appear only as Watchlist and must be clearly labeled.
- Favor securities with reliable Massive daily bars, current-session data, and enough options/news data to validate the setup. Missing optional data does not automatically reject a stock, but missing hard-gate data must remain visible.

Technical setup & momentum gates

- Calculate 1-day, 5-day, 10-day, and 20-day returns; 10-day and 20-day moving averages; 20-day high; 10-day low; daily close location; and relative volume versus the prior 20 sessions.
- Prefer positive 5-day momentum, price above both the 10-day and 20-day moving averages, and a close within approximately 3% of the 20-day high or a clean breakout through that high.
- Prefer RSI(14) roughly 50-68. RSI 68-75 may still qualify when volume/catalyst support is strong; RSI >75 is normally Watchlist unless a fresh catalyst materially changes the setup.
- Reward strong closes: closing in the upper portion of the daily range and near the session high. Penalize late-day reversals, wide upper wicks, and closes materially below VWAP when current-session VWAP is available.
- Reward relative volume $\geq 1.2x$ the recent norm. Volume below $0.8x$ weakens a breakout unless another catalyst provides clear confirmation.

Historical 3-5 day persistence test

- Use at least approximately 252 daily bars when available. Reconstruct prior sessions that resemble the current setup using only information available as of each historical signal date.
- For comparable historical signals, calculate forward 3-day and 5-day close returns, probability of finishing positive, average and median returns, +1% and +2% touch rates, worst return, and preferably the 5th-percentile return.
- Full Pass should normally show $\geq 60\%$ 5-day positive rate with positive average and median forward returns, subject to a reasonable sample size. If sample size is thin or recent persistence materially deteriorates, classify as Watchlist rather than claiming a high-probability edge.

- Historical statistics are supporting evidence, not a forecast. A stock can still fail because of gaps, news, macro events, or regime changes.

Options positioning & catalyst confirmation

- When an options chain is available, compare near-term call versus put volume and open interest, focusing on liquid strikes around spot and the next one or two practical expirations.
- Call/put volume >1.5x is supportive; >2.5x is strong confirmation when the activity is liquid and not obviously distorted by one isolated contract. Options flow is a confirmation factor, not a stand-alone buy signal.
- Review the most recent company news and Massive sentiment/insight fields where available. Reward fresh positive catalysts such as earnings/guidance beats, index additions, analyst estimate revisions, product or contract developments, shareholder-return actions, or sector-specific tailwinds.
- Do not treat promotional commentary or a single opinion article as equivalent to a primary company catalyst. When news is mixed, state the conflict explicitly.

Market, sector & event-risk controls

- Compare the candidate with SPY/QQQ and its relevant sector or industry ETF when practical. Reward relative strength when the stock is outperforming a weak or flat broad market.
- Identify confirmed or projected earnings inside the intended 3-5 trading-day holding window. By default, earnings inside the window prevent Full Pass unless the user explicitly asks for an event trade.
- Flag known binary company events and major macro events inside the holding window, including FOMC decisions, CPI, payrolls, or other releases likely to overwhelm the technical setup.
- If a major market-moving event falls inside the window, reduce conviction, prefer smaller targets/earlier profit-taking, and clearly label the event date/time.

Ranking score

Component	Approx. Weight
Trend / multi-day momentum	25%
Breakout quality / close strength	15%
Relative volume / liquidity	15%
Historical 3-5 day persistence / downside tail	15%
Options positioning confirmation	10%
Fresh catalyst / news quality	10%
Sector / market relative strength	5%
Extension / gap / RSI risk	5%

Classification & execution framing

- Full Pass / A-range: score ≥ 75 with no hard event veto, adequate liquidity, and no unresolved data problem that would materially change the ranking.
- Watchlist / B-range: score 65-74, or an otherwise strong setup with a material caveat such as overextension, thin historical sample, conflicting news, unverified earnings timing, or a major macro event inside the hold window.

- Reject: score <65 or failure of a hard gate such as unacceptable liquidity, clearly broken trend, imminent earnings/event risk under the default rules, or severe overextension without catalyst support.
- Do not automatically buy the opening print. For premarket or opening-bell scans, prefer confirmation during roughly the first 15-30 minutes unless the user specifically requests another execution style.
- If the stock gaps >3% above the prior close, recalculate the setup before entry. A gap >5% normally triggers a no-chase warning unless the new price still offers favorable support/target asymmetry.

Required output fields

Rank; ticker/company; setup grade/score; current or prior close; current day/premarket context when available; 1/5/10/20-day returns; RSI(14); 10-day/20-day moving averages; distance to 20-day high; relative volume; close-location / VWAP context; 3-day and 5-day historical win statistics and downside-tail metrics when sample permits; options call/put volume and OI confirmation when available; fresh catalyst/news summary; sector/market relative strength; earnings date/status; major event-risk flags; preferred entry zone; breakout trigger; 3-5 day target zone; invalidation level; reason for rank; Full Pass / Watchlist / Reject; and an explicit no-chase note when applicable.

Important: The scanner estimates relative setup quality; it does not guarantee that a stock will rise over the next 3-5 trading days. Verify live pricing and any material catalyst before trading.

9. 90-Day / 15% Downside Safety Scanner

Purpose and task status

This scanner identifies liquid U.S.-listed common stocks and conventional ETFs whose history shows a very high frequency of avoiding a path drawdown greater than 15% during the following 63 trading sessions. It is designed as a downside-risk screen and as a research layer for conservative put credit spreads; it is not a guarantee that a future breach cannot occur.

Setting	Current specification
Task status	Paused scheduled task; manual on-demand runs remain available
Schedule when enabled	Monday-Friday at 8:30 AM PT; skip U.S. market holidays
Primary data	Massive adjusted daily OHLCV, reference data, corporate actions, and options data
Instruments	Common stocks and conventional ETFs; optional put credit spread overlay
Email	Send every completed report to Dennis Weng at dwengca@gmail.com

Universe and liquidity

- Include active U.S.-listed common stocks and conventional broad-market, index, sector, industry, factor, international, bond, and commodity ETFs when all gates pass.
- Exclude OTC securities, preferreds, warrants, rights, units, closed-end funds, non-primary ADRs, leveraged or inverse ETFs, volatility/VIX products, single-stock ETFs, synthetic high-leverage products, ETNs, unresolved corporate actions, and materially incomplete histories.

- Require current price from \$5 to \$1,000, prior-30-session average share volume of at least 1.5 million, average daily dollar volume of at least \$100 million, and listed options.
- For a displayed spread, require at least 500 contracts of open interest on each leg. Reject stale, crossed, zero-bid, or distorted option markets.

Historical downside test

Use about 10 years of split- and distribution-adjusted daily data when available, with a hard minimum of five years. For each eligible historical start session t , set $P(t)$ to that session's adjusted close and $Lmin(t)$ to the lowest adjusted daily low during sessions $t+1$ through $t+63$.

$$MAE63(t) = Lmin(t) / P(t) - 1$$

- A trial survives only when MAE63 is greater than -15%. A touch of exactly -15% counts as a breach.
- The test is path-dependent. A security fails when it breaches the threshold at any point, even if it recovers by session 63.
- Primary survival rate equals surviving trials divided by all eligible trials. Label it an historical frequency estimate, not a guaranteed probability.

Robustness and classification

- Report full-history and recent-five-year survival rates; a non-overlapping quarterly-start rate; and a 95% Wilson lower confidence bound for the non-overlapping sample.
- When available, report survival by VIX regime below 20, 20-30, and above 30, plus results when the underlying starts below its 200-day moving average.
- Report worst observed MAE and date, breach count, median breach, 5th-percentile MAE, and outcomes during available stressed periods including 2020 and 2022.

Classification	Requirement
Tier A - Full Pass	Full history $\geq 95\%$; recent five years $\geq 95\%$; non-overlapping Wilson lower bound $\geq 90\%$; all hard gates pass
Tier B - Strong Watchlist	Full history and recent five years $\geq 90\%$, but one Tier A condition is missed
Tier C - Watchlist	Full-history rate $\geq 85\%$ with no unresolved data-quality problem
Reject	Full history $< 85\%$, inadequate history, failed liquidity/options gate, excluded product, or unreliable data

Current-risk and options overlay

- Report current price, the current 15% downside threshold, day change, 20/50/200-day moving averages, RSI(14), ATR(14) as a percentage of price, 52-week high/low, current VIX regime, and the closest-regime historical survival rate.
- For stocks, report the next confirmed earnings date. A stock cannot receive options-trade Full Pass when earnings occur on or before expiration or within seven calendar days after expiration. ETFs are marked N/A - ETF for earnings.
- Search 30-120 DTE and prioritize 75-105 DTE. The short strike must be at or below 85% of spot; designate it preferred when it also satisfies the baseline limits of no more than 80% of spot and no more than 120% of the 52-week low.

- Use net spread delta = (-short-put delta) + long-put delta. Tier A ≤ 0.03 ; Tier B > 0.03 to 0.05 ; Tier C > 0.05 to 0.08 Watchlist; reject > 0.08 .
- Require estimated credit $\geq \$0.25$ and $\text{RoR} = \text{credit} / (\text{width} - \text{credit}) \geq 5\%$. Report strikes, width, expiration/DTE, deltas, IV, OI, volume, bid/ask timestamp, credit, max risk, RoR, breakeven, and distance to the short strike.
- Treat option quotes as screening indications and verify live executable prices in Robinhood before any trade.

Output and email delivery

- Return Data Status and Methodology; Tier A Full Pass; Tier B Strong Watchlist; Tier C and Near-Misses; Reject and Market Context; optional put-credit-spread candidates; and Bottom Line.
- For every reported underlying include type, price, 15% threshold, history dates, trial counts, survival rates, Wilson lower bound, worst MAE/date, 5th-percentile MAE, crisis results, current-risk metrics, options-liquidity status, and classification reason.
- After every completed run, email the complete report to Dennis Weng at dwengca@gmail.com with subject '90-Day Downside Safety Scanner - [run date in America/Los_Angeles]'. Send a report even when no Tier A candidates exist; include the strongest near-misses and failure reasons.
- If email delivery fails, preserve the full task response and clearly report the failure. Do not claim delivery unless Gmail confirms success.

Risk note: A high historical survival frequency does not estimate the probability of profit for a specific option spread and cannot eliminate gap, regime, liquidity, event, or execution risk. ETF diversification does not eliminate market, sector, duration, currency, commodity, or structural risk.

10. Scanner Comparison Matrix

Rule	Put Spread	Put Skew	Weak Close	5-Day Weak Close	Friday
Schedule	M-F 8:30	M-F 8:30	M-F 12:40	M-F 12:40	Fri 12:40
Market-cap floor	None explicit; prioritize large cap	Generally $\geq \$10B$	$\geq \$10B$	$\geq \$10B$	$\geq \$10B$
Stock volume floor	$\geq 1.5M$ sh	$\geq 1.5M$ sh	$\geq 1M$ sh + \$100M ADV	$\geq 1M$ sh + \$100M ADV	$\geq 1M$ sh + \$100M ADV
DTE / hold	30-120 DTE	30-120 DTE	Overnight	≤ 5 trading days	Weekend to Monday
Primary edge	Deep OTM + low delta	Downside IV skew	Next-open rebound	3-5 day rebound	Monday rebound
Earnings rule	Expiry ≥ 7 days before earnings	Expiry ≥ 7 days before earnings	No earnings within 7 days	No earnings within 7 days / holding window	No weekend / Mon-pre-exit earnings
Full-pass sample	N/A	N/A	≥ 40 weak closes	≥ 40 weak closes	≥ 25 Fridays
Key win threshold	Economics + delta/OI gates	Skew + economics + delta	Next-open $\geq 60\%$	5-day $\geq 65\%$	Mon open & close $\geq 60\%$
Recent persistence	Market/chain current	Market/chain current	$\sim 60d \geq 60\%$	$\sim 60d \geq 60\% + \text{avg} > 0$	$\sim 12m \geq 60\% + \text{avg} > 0$

On-demand scanner comparison addendum

Rule	3-5 Day Swing
Schedule	On demand
Market-cap floor	No hard floor; liquidity gates dominate
Stock volume floor	Prefer ≥ 1 M shares + \$100M 30-day ADV
DTE / hold	3-5 trading days
Primary edge	Momentum + breakout + catalyst continuation
Earnings rule	No earnings inside hold window by default
Full-pass sample	Historical comparable-signal sample should be reasonable; thin sample -> Watchlist
Key win threshold	Normally $\geq 60\%$ 5-day positive rate + positive avg/median, plus score ≥ 75
Recent persistence	Must not materially contradict longer-history signal

Paused scanner comparison addendum

Rule	90-Day / 15% Downside Safety
Schedule	M-F 8:30 AM PT when enabled; currently paused; manual runs allowed
Instrument	Common stock or conventional ETF; optional put credit spread
Liquidity floor	≥ 1.5 M shares and $\geq \$100$ M average daily dollar volume
History	About 10 years preferred; five years minimum
Horizon	63 trading sessions, approximately 90 calendar days
Primary edge	Historical avoidance of a $>15\%$ path drawdown
Tier A threshold	$\geq 95\%$ full-history and recent-five-year survival; Wilson lower bound $\geq 90\%$
Event rule	Stock expiration must clear confirmed earnings by at least seven calendar days
Delivery	Complete result emailed to dwengca@gmail.com after every run

11. Massive Data Implementation Guide

This section explains how to reproduce the scanner data pipeline with Massive. It supplements the trading rules above; it does not replace them. Massive plan entitlements can change, so confirm endpoint access and data recency against the current Massive documentation before deployment.

Authentication & API hygiene

- Use a Massive API key stored as an environment variable or secret manager. Do not publish, hard-code, or embed the key in this playbook or source code.

- Base REST URL: <https://api.massive.com>. Authenticate using the Massive client library or the API key mechanism documented by Massive.
- Implement retries with backoff for transient errors, log request IDs, and cache reference/history data so a full-universe scan does not repeatedly download unchanged data.
- When an endpoint returns `next_url`, continue paging until no `next_url` remains. Option-chain pages can contain up to 250 results, so never assume the first response is the complete chain.
- Timestamp every scan and record whether stock/options observations are real-time, delayed, end-of-day, or indicative. Do not mix observations with different recency without labeling them.

Core Massive endpoints and fields

Need	Massive source	Scanner use
Stock current/session data	Stock snapshot / unified snapshot	Current price, regular-session open/high/low/close, previous close, volume; use for current-day change and close-location.
Stock historical bars	Stock aggregate bars (daily)	Daily OHLCV; use for 30-day volume, dollar volume, 52-week low, realized volatility, weak-close history and forward-return tests.
Options contract universe	GET <code>/v3/reference/options/contracts</code>	Underlying, contract type, expiration, strike, option ticker; filter to puts and the target 30-120 DTE window.
Options chain snapshot	GET <code>/v3/snapshot/options/{underlying Asset}</code>	Per-contract details, Greeks, implied volatility, open interest, underlying data and quotes/trades when the plan provides them.
Single option snapshot	GET <code>/v3/snapshot/options/{underlying Asset}/{optionContract}</code>	Use to refresh or verify an individual short/long leg after the chain has been narrowed.
Unified snapshot	GET <code>/v3/snapshot</code>	Optional multi-asset snapshot path for current stock/options observations where plan access permits.

Canonical stock-data workflow

1. Build the eligible common-stock universe first. Apply the scanner-specific market-cap, security-type, price, and liquidity exclusions before making expensive option-chain calls.
2. Fetch daily stock bars for enough history to cover the calculation: at least ~252 sessions for Weak Close scanners and the Swing Scanner historical-persistence test, approximately 2 years for Friday, and at least 52 weeks when the 52-week low is required, and about 10 years (five years minimum) for the 90-Day Downside Scanner.

3. Compute daily return = close / prior close - 1; dollar volume = close × volume; 30-day average share/dollar volume; 52-week low; and RV30 from daily log or simple returns using one consistent annualization convention.
4. For late-day scanners, obtain the current regular-session open/high/low/current price and compute close-location = (current - low)/(high - low). If high equals low, mark the observation invalid rather than dividing by zero.
5. Use official regular-session daily bars for historical backtests. Do not substitute pre-market or after-hours prints for the official open/close statistics.

Canonical options workflow

6. For each stock surviving the stock gates, request put contracts/chain data and restrict expirations to 30-120 DTE.
7. Apply strike-distance rules before pairing legs: Put Spread short strike $\leq 80\%$ of spot and $\leq 120\%$ of the 52-week low; Put Skew short strike generally $\leq 85\%$ of spot with short-put delta approximately -0.05 to -0.15.
8. Require the scanner-specified open interest. Put Skew requires ≥ 500 on both legs; Put Spread reports both legs and requires qualifying near-strike liquidity.
9. Pair the short higher-strike put with a lower-strike long put using a practical width. Calculate net spread delta = $(-1 \times \text{short-put delta}) + \text{long-put delta}$; when both put deltas are negative this equals $|\text{short delta}| - |\text{long delta}|$.
10. Calculate spread credit from executable-quality quotes when available: conservative credit estimate = short bid - long ask. Midpoint estimates may be displayed separately but must not be represented as executable.
11. Calculate max risk = spread width - net credit and RoR = net credit / max risk. Apply the \$0.25 minimum credit and 5% RoR Full Pass gates.
12. For Put Skew, estimate ATM IV from the closest-to-ATM liquid options, then compute skew premium = short-put IV - ATM IV and IV/RV = short-put IV / RV30.
13. If Greeks, IV, OI, or quote fields are missing, do not synthesize them. Mark the candidate incomplete, use Watchlist where the live scanner rules permit, or reject when the field is a hard gate.

Scanner-to-data mapping

Put Spread: Stock bars/snapshot provide day change, 30-day volatility behavior, volume and 52-week low. The options chain provides expiration, strike, Greeks, IV and OI. Massive is the primary quantitative source; the active scanner may use Alpaca for secondary indicative bid/ask economics, followed by live Robinhood verification.

Put Skew: Use stock bars for RV30 and liquidity. Use the options chain for ATM IV, short-put IV, individual deltas, OI and available quotes. Calculate skew premium, IV/RV and net spread delta locally.

Weak Close: Use current stock session data near 12:40 PM PT and daily bars for ~252 sessions. Reconstruct historical weak closes and the next official regular-session open. Calculate sample count, win rate, mean, median and downside-tail statistics locally.

5-Day Weak Close: Use the same current/history stock data as Weak Close, but calculate 3-day/5-day forward closes and whether subsequent daily highs touch +1% or +2% during the five-session window.

Friday: Use Friday current-session data near 12:40 PM PT and approximately two years of daily bars. Identify historical weak Fridays, map each to the next normal Monday session, and calculate Monday

open/close returns, hit rates and downside statistics. Holiday weekends whose next session is Tuesday must be separated.

3-5 Day Swing: Use stock daily bars and current/session snapshots for momentum, moving averages, breakout proximity, close strength, VWAP context and relative volume. Use RSI or locally calculated RSI(14). Use option-chain snapshots for call/put volume, OI, IV and Greeks when available. Use Massive news/sentiment for catalyst confirmation, and compare against broad-market/sector proxies. Reconstruct comparable historical signals without look-ahead and measure forward 3-day/5-day returns and downside tails.

90-Day Downside: Use adjusted stock/ETF daily bars for approximately 10 years when available, minimum five years. For each start date, compare its adjusted close with the lowest adjusted low in the next 63 sessions. Calculate overlapping and non-overlapping survival rates, a Wilson lower bound, downside-tail statistics, VIX-regime results, and below-200-day-moving-average results. Use option-chain snapshots only after the underlying passes the historical and liquidity gates.

Earnings and event data

Earnings is a hard safety dependency, but it should not be inferred from option IV or price action. Use Massive earnings/corporate-event data when the connected plan exposes a reliable date/time. Otherwise verify with a reliable current earnings source, preferring the issuer's investor-relations calendar. Store the source, retrieval timestamp, confirmed/projected status, and event time (before open / after close / unknown). If the scanner requires earnings timing and it cannot be verified, do not award Full Pass.

Data-quality and reproducibility rules

- Save the scanner version, run timestamp, data recency, source, and all hard-gate inputs with every result.
- Use split-adjusted historical stock data consistently for return tests and 52-week comparisons.
- Never silently fill missing bars, Greeks, IV, OI, earnings dates, or quotes. Missing hard-gate data must remain visible in the classification.
- Backtests must avoid look-ahead: a historical signal may use only information available as of that historical close.
- Use the next actual trading session rather than calendar-day arithmetic for forward returns, and treat Friday holiday exceptions exactly as specified in the Friday scanner.
- Before publication or sharing, remove API keys, account identifiers, broker credentials, and any private connection details.

Minimal implementation sequence

14. Load eligible tickers and reference attributes.
15. Fetch/cache stock history and compute common stock metrics.
16. Apply each scanner's inexpensive stock gates.
17. For Put Spread / Put Skew survivors only, request and fully paginate the Massive options chain.
18. Construct candidate verticals and calculate local derived metrics (net delta, skew, max risk, RoR).
19. Apply earnings/event vetoes and classification rules.
20. Rank Full Pass first, then Watchlist/near-misses, and emit every required field defined in the scanner specification.
21. For options trades, verify the final spread in Robinhood immediately before a trade decision; published scanner output is screening information, not an executable quote.

Massive reference documentation

- Options overview: <https://massive.com/docs/rest/options/overview>
- Option chain snapshot: <https://massive.com/docs/rest/options/snapshots/option-chain-snapshot>
- Option contract snapshot: <https://massive.com/docs/rest/options/snapshots/option-contract-snapshot>
- Options contracts: <https://massive.com/docs/rest/options/contracts>
- Stock snapshots: <https://massive.com/docs/rest/stocks/snapshots>

12. Publication Notes

This document is a publishable specification of the active, paused, and on-demand scanner logic as of September 16, 2026. It is intended to make the scanners transparent, repeatable, and shareable without exposing private credentials or account identifiers.

- Times are Pacific Time unless otherwise stated.
- “Friday” is the active task name for the Friday → Monday Effect Scanner.
- The former Weekday Scanner is not part of this playbook.
- When scanner logic or task status changes, update the version/date and revise the affected scanner section so the published document remains aligned with the live scheduled task.

Risk notice: Historical win rates, average returns, implied volatility relationships, and low delta do not guarantee future outcomes. Options carry assignment, gap, liquidity, and loss risk; equities can gap through historical patterns on new information.

- The 90-Day / 15% Downside Safety Scanner is stored as a paused weekday task. It can be run manually on demand and sends each completed result to dwengca@gmail.com.
- The 3-5 Day Swing Scanner is on demand only and does not consume a scheduled-task slot. Default trigger phrase: “Run my 3-5 Day Swing Scanner using Massive.”

PUBLIC SHARING NOTICE: USE AT YOUR OWN RISK. Verify every scanner result and every proposed trade independently before acting. No scanner can eliminate market, event, data, model, liquidity, execution, or assignment risk.

Appendix A - Scanner Prompts (Copy & Paste)

The following prompts mirror the active scheduled scanners, the paused 90-Day / 15% Downside Safety Scanner, and the on-demand 3-5 Day Swing Scanner as of September 16, 2026. They are included so readers can recreate the scanner logic. Data access, connected services, and scheduling must be configured separately.

Put Spread Scanner

Run Dennis's daily put-credit-spread stock filter using the connected Massive market-data plugin as the primary source and connected Alpaca as a secondary indicative-pricing source. Scan liquid U.S. equities with stock prices from \$3 to \$1,000, prioritizing large-cap names and S&P 500/Nasdaq-100 when appropriate. Gate 1: stock is currently down on the day; over the last 30 trading days, at least 5 days had absolute daily returns >2%; no single daily decline was <=-4%; average daily volume >=1.5M shares. Gate 2: target 30-120 DTE; short put strike must be <=80% of current price and <=120% of the 52-week low. DELTA METHOD: use NET PUT-SPREAD DELTA, calculated from Massive individual-leg Greeks for each candidate spread. For a standard put credit spread that is short the higher-strike put and long the lower-strike put, calculate position net delta as $(-1 \times \text{short-put delta}) + \text{long-put delta}$; equivalently, when both put deltas are negative, $\text{abs}(\text{short-put delta}) - \text{abs}(\text{long-put delta})$. Use the absolute magnitude of this net spread delta for screening and tier classification. Delta tiers: Tier A Ideal = net spread absolute delta <=0.03; Tier B Acceptable = >0.03 and <=0.05; Tier C Watchlist only = >0.05 and <=0.08; reject net spread delta >0.08. Do not use the individual short-put delta for the A/B/C gate. However, always report the individual short-put delta and long-put delta as secondary risk metrics alongside the calculated net spread delta so results can be compared with Robinhood's displayed spread delta. Do not treat Tier C as a full pass. EARNINGS EVENT FILTER: determine the next expected or confirmed earnings date for each candidate using a reliable current earnings-calendar source; use Massive earnings data if accessible under the connected plan, otherwise use reliable current web sources and prefer company investor-relations information when available. A FULL PASS spread must expire at least 7 calendar days BEFORE the next earnings event. If the next earnings date occurs on or before 7 calendar days after the option expiration, reject that expiration from the Full Pass list and look for another qualifying expiration within 30-120 DTE that clears the earnings buffer. If no qualifying pre-earnings expiration exists, exclude the stock from Full Pass. If the earnings date is only projected/unconfirmed, clearly label it PROJECTED and still apply the 7-day buffer conservatively. If the next earnings date cannot be established reliably, do not award Full Pass; place an otherwise qualifying candidate in the Watchlist with 'EARNINGS DATE UNVERIFIED'. Report the next earnings date, whether it is confirmed or projected, and the number of calendar days between option expiration and earnings. Use Massive full options-chain snapshots for expiration, strike, individual-leg delta, IV, and open interest, and calculate net spread delta from the two leg deltas. Require near-strike option open interest >=500 contracts. Use a spread width appropriate to the stock price and available strikes, generally \$5-wide for higher-priced liquid stocks, with \$2.50, \$1, or other practical widths for lower-priced stocks. For contracts that survive these hard screening gates, use Alpaca's connected indicative options feed to obtain indicative bid/ask for both legs and estimate net credit and return on risk. Clearly label all Alpaca-derived bid/ask, net credit, and RoR as INDICATIVE/ESTIMATED, not executable quotes. Use minimum estimated net credit >=\$0.25 and estimated RoR (net credit / (spread width - net credit)) >=5% as screening thresholds. Treat combined indicative bid-ask width <=\$0.08 as a preferred liquidity screen, but if indicative data are too sparse or distorted to apply it reliably, flag the candidate rather than inventing data. Never claim the estimated credit is currently executable. For every candidate that otherwise passes, include a prominent 'VERIFY IN ROBINHOOD BEFORE TRADE' note requiring the live Robinhood spread credit to be >=\$0.25 and live RoR >=5%; if the live Robinhood economics fail, do not trade based on the screening result. Rank candidates by overall quality, emphasizing downside buffer, net spread delta tier, earnings-event clearance, Massive OI/liquidity, estimated credit, and estimated RoR. For each candidate, report ticker/company, spot price and daily change, 52-week low, expiration/DTE, next earnings date and confirmed/projected status, days from expiration to earnings, short and long strikes, individual short-put delta, individual long-put delta, calculated net spread delta and tier, indicative bid/ask on both legs when available, estimated net credit, open interest on both legs, IV if available, max risk based on estimated credit, estimated RoR, downside buffer to short strike, 30-day volatility check, and key reasons it passed. Exclude candidates that fail any hard Gate 1, Gate 2, earnings-buffer, net-spread-delta-tier, or OI requirements. If useful, provide a separate small Watchlist for Tier C candidates or otherwise qualifying candidates whose earnings date or pricing requires verification. If Massive or Alpaca data are incomplete, explicitly flag the missing data rather than substituting generic web-scraped options data. This is a screening list for consideration, not an instruction to trade; do not claim any trade is risk-free.

Use at your own risk. Verify all results, data freshness, earnings timing, liquidity, and live broker pricing before acting.

Put Skew Scanner

Run the Put Skew Scanner using Massive as the primary stock/options data source. Goal: find liquid U.S. common stocks where out-of-the-money downside puts are unusually expensive relative to ATM implied volatility and recent realized volatility, then identify conservative put credit spreads that monetize the skew. Universe: active U.S.-listed common stocks, prioritize large-cap / S&P 500 / Nasdaq 100, generally market cap $\geq$ \$10B where reference data is available; exclude ETFs, leveraged/inverse funds, OTC, preferreds, warrants, and non-common shares. Stock liquidity: average share volume over the prior 30 trading days $\geq$ 1.5M. Expirations: 30-120 DTE. Earnings/event filter: for Full Pass, expiration must be at least 7 calendar days BEFORE the next confirmed or conservatively projected earnings event; if expiration crosses earnings, search another qualifying expiration. If earnings date cannot be verified, classify as Watchlist with EARNINGS DATE UNVERIFIED. Flag known material/binary events before expiration. For each candidate expiration, estimate ATM IV using the closest-to-ATM liquid options. Search downside put spreads with short-put delta approximately -0.05 to -0.15, short strike generally $\leq$ 85% of underlying price, and practical spread width generally \$5 for higher-priced stocks and \$2.50 or \$1 for lower-priced stocks. Require open interest $\geq$ 500 contracts on BOTH legs for Full Pass. Compute realized volatility from recent stock returns, emphasizing 30-day realized volatility (RV30). Compute Put Skew Premium = short-put IV - ATM IV. Compute IV/RV ratio = short-put IV / RV30. Compute net spread delta using NET PUT-SPREAD DELTA, not short-leg delta: net delta = $(-1 \times \text{short-put delta}) + \text{long-put delta} = \text{abs}(\text{short delta}) - \text{abs}(\text{long delta})$ when both puts are negative. Always report individual short- and long-put deltas as secondary metrics. Net delta tiers: Tier A Ideal $\leq$ 0.03; Tier B > 0.03 and $\leq$ 0.05; Tier C Watchlist > 0.05 and $\leq$ 0.08; Reject > 0.08 . Tier A skew target: short-put IV at least 8 volatility points above ATM IV AND IV/RV $\geq$ 1.25. Tier B skew target: skew premium 5-8 volatility points AND IV/RV $\geq$ 1.15. Candidates below these skew thresholds should generally be Watchlist or Reject depending on overall quality. Pricing: use available live/indicative bid/ask data to estimate net credit and return on risk; label any non-live or stale pricing as INDICATIVE/ESTIMATED. Full Pass requires estimated/live net credit $\geq$ \$0.25 and RoR $\geq$ 5%. Prefer combined spread bid-ask width $\leq$ \$0.08; flag sparse or wide pricing. Rank candidates approximately by 30% skew premium, 20% IV/RV premium, 20% downside distance to short strike, 15% spread RoR, 10% liquidity/OI, and 5% net delta, then apply earnings/event vetoes. Penalize unusually poor liquidity, extremely elevated event-driven IV, or severe underlying downside tails that suggest the skew may reflect genuine binary risk. For each result report: ticker/company, underlying price, expiration and DTE, short/long strikes, short strike as % of spot, short/long IV, ATM IV, skew premium in vol points, RV30, IV/RV ratio, short delta, long delta, net spread delta and tier, OI each leg, bid/ask each leg if available, estimated net credit, max risk, RoR, earnings date/status/days to earnings, event-risk notes, and concise reason for ranking. Classify as Full Pass / Watchlist / Reject. Always state VERIFY IN ROBINHOOD BEFORE TRADE and that this is a statistical screening list, not a guarantee of safety or profit. If no Full Pass candidates exist, show the best near-misses and exactly which criteria they failed.

Use at your own risk. Verify all results, data freshness, earnings timing, liquidity, and live broker pricing before acting.

Weak Close Scanner

Run Dennis's Weak Close Scanner for U.S.-listed common stocks. Goal: identify liquid mid-cap to large-cap stocks showing a historically persistent weak-close-to-next-open rebound pattern suitable for consideration as a buy-near-close / sell-next-morning setup. Use connected Massive market data as the primary quantitative source. Universe: active U.S.-listed common stocks with current market capitalization $\geq$ \$10 billion; exclude ETFs, leveraged/inverse products, OTC securities, preferreds, warrants, and other non-common-stock instruments. Require strong liquidity: average daily dollar volume $\geq$ \$100 million over the prior 30 trading days and average daily share volume $\geq$ 1 million. CURRENT-DAY SETUP: evaluate near the end of the regular session. Candidate must be trading below today's regular-session open and in the bottom 30% of today's regular-session high-low range, using close-location = $(\text{current price} - \text{day low}) / (\text{day high} - \text{day low}) \leq 0.30$. Prefer meaningful weakness rather than tiny noise; report today's open-to-current return and close-location percentile. HISTORICAL PATTERN TEST: use approximately the prior 252 trading days. For each historical day, define a weak close as regular-session close $<$ regular-session open AND close-location = $(\text{close-low}) / (\text{high-low}) \leq 0.30$. For those historical weak-close occurrences, calculate the next regular-session opening return = $(\text{next trading day's official regular-session open} / \text{prior close} - 1)$. Require at least 40 historical qualifying weak-close observations. Full Pass requires next-open win rate $\geq$ 60%, average next-open return $> 0\%$, and median next-open return $> 0\%$. RECENT PERSISTENCE TEST: independently evaluate approximately the most recent 60 trading days using the same weak-close definition. Require at least 10 recent qualifying observations and

next-open win rate $\geq 60\%$; report recent average overnight return. EARNINGS SAFETY FILTER: exclude any stock with a confirmed or projected earnings event within the next 7 calendar days, including tonight/tomorrow. Determine the next earnings date using a reliable current earnings-calendar source; use Massive earnings data when accessible and otherwise use reliable current web sources, preferring company investor-relations information. If the earnings date cannot be established reliably, do not award Full Pass; place an otherwise qualifying stock in Watchlist with 'EARNINGS DATE UNVERIFIED'. EVENT RISK: flag known scheduled material company events or other clearly identifiable binary events before the next regular-session open; do not claim all unscheduled overnight news risk can be eliminated. RANKING: rank Full Pass candidates primarily by historical next-open win rate, recent persistence, median and average overnight return, sample size, liquidity, and downside-tail behavior. Penalize unstable patterns dominated by a few large gaps. REPORT for each candidate: ticker/company, market cap, current price, today's open/high/low, open-to-current % move, current close-location %, 1-year weak-close sample count, 1-year next-open win rate, average and median next-open return, worst overnight return and preferably 5th-percentile overnight return, recent ~60-day sample count/win rate/average return, next earnings date and confirmed/projected status, days until earnings, liquidity metrics, and key reason it qualified. Clearly distinguish Full Pass, Watchlist, and Reject. This is a statistical screening tool, not a prediction that a stock will open higher. Do not describe any setup as nearly guaranteed or risk-free. Overnight gaps can cause losses larger than the expected average gain. Run this scan at approximately 3:40 PM Eastern Time on regular U.S. trading days so the current-day weak-close condition can be evaluated before the 4:00 PM close.

Use at your own risk. Verify all results, data freshness, earnings timing, liquidity, and live broker pricing before acting.

5-Day Weak Close Scanner

Run Dennis's 5-Day Weak Close Rebound Scanner for U.S.-listed common stocks. Goal: identify liquid mid-cap to large-cap stocks that historically tend to rebound over the next several trading days after a weak close, for consideration as a buy-near-close / exit-within-5-trading-days setup. Use connected Massive market data as the primary quantitative source. Universe: active U.S.-listed common stocks with current market capitalization $\geq \$10$ billion; exclude ETFs, leveraged/inverse products, OTC securities, preferreds, warrants, and other non-common-stock instruments. Require strong liquidity: average daily dollar volume $\geq \$100$ million over the prior 30 trading days and average daily share volume ≥ 1 million. CURRENT-DAY SETUP: evaluate near the end of the regular session. Candidate must be trading below today's regular-session open and in the bottom 30% of today's regular-session high-low range, using close-location = $(\text{current price} - \text{day low}) / (\text{day high} - \text{day low}) \leq 0.30$. Prefer meaningful weakness rather than tiny noise; report today's open-to-current return and close-location percentile. HISTORICAL PATTERN TEST: use approximately the prior 252 trading days. For each historical day, define a weak close as regular-session close $<$ regular-session open AND close-location = $(\text{close-low}) / (\text{high-low}) \leq 0.30$. For each historical weak-close occurrence, calculate forward returns from that close to the regular-session close 3 trading days later and 5 trading days later. Also calculate whether the stock touched +1% and +2% at any point during the next 5 trading days using the subsequent daily highs versus the weak-close entry price. Require at least 40 historical qualifying weak-close observations. FULL PASS thresholds: 5-trading-day win rate $\geq 65\%$; average 5-day return $> 0\%$; median 5-day return $> 0\%$; probability of touching +1% within the next 5 trading days $\geq 80\%$. Report 3-day win rate, 5-day win rate, average and median 5-day return, +1% hit rate, +2% hit rate, worst 5-day return, and preferably 5th-percentile 5-day return. RECENT PERSISTENCE TEST: independently evaluate approximately the most recent 60 trading days using the same weak-close definition. Require at least 10 recent qualifying observations, recent 5-day win rate $\geq 60\%$, and recent average 5-day return $> 0\%$. If historical Full Pass thresholds are met but recent persistence fails, classify as Watchlist rather than Full Pass. EARNINGS SAFETY FILTER: exclude any stock with a confirmed or projected earnings event within the next 7 calendar days, including tonight/tomorrow. Because this strategy may hold up to 5 trading days, if earnings falls before the end of that expected holding window plus a safety buffer, do not award Full Pass. Determine the next earnings date using a reliable current earnings-calendar source; use Massive earnings data when accessible and otherwise reliable current web sources, preferring company investor-relations information. If the earnings date cannot be established reliably, do not award Full Pass; place an otherwise qualifying stock in Watchlist with 'EARNINGS DATE UNVERIFIED'. EVENT RISK: flag known scheduled material company events or clearly identifiable binary events occurring during the next 5 trading days; do not claim all unscheduled news risk can be eliminated. TAIL-RISK CONTROL: penalize unstable patterns dominated by a few large rebounds or containing severe downside outliers. Report worst 5-day return and 5th-percentile 5-day return, and demote candidates with unusually poor downside tails even if headline win rate is high. RANKING: rank Full Pass candidates primarily by 5-day win rate, recent persistence, median and average 5-day return, +1% hit rate, sample size, liquidity, and downside-tail behavior. REPORT for each candidate: ticker/company, market cap, current price, today's open/high/low, open-to-current % move, current close-

location %, 1-year weak-close sample count, 3-day win rate, 5-day win rate, average and median 5-day return, +1% hit rate within 5 days, +2% hit rate within 5 days, worst 5-day return, 5th-percentile 5-day return if available, recent ~60-day sample count/5-day win rate/average 5-day return, next earnings date and confirmed/projected status, days until earnings, liquidity metrics, and key reason it qualified. Clearly distinguish Full Pass, Watchlist, and Reject. This is a statistical screening tool, not a prediction that a stock will be higher in 5 days. Do not describe any setup as nearly guaranteed or risk-free. Multi-day gaps and news can cause losses larger than the expected average gain. Run this scan at approximately 3:40 PM Eastern Time on regular U.S. trading days so the current-day weak-close condition can be evaluated before the 4:00 PM close.

Use at your own risk. Verify all results, data freshness, earnings timing, liquidity, and live broker pricing before acting.

Friday - Friday to Monday Effect Scanner

Run the Friday -> Monday Effect Scanner using Massive as the primary market-data source. Goal: identify liquid U.S.-listed common stocks late on Friday that historically tend to rebound by Monday after a weak Friday close, for consideration as a buy-near-Friday-close / exit-Monday setup. Universe: active U.S.-listed common stocks with market capitalization $\geq$ \$10 billion; exclude ETFs, leveraged/inverse products, OTC securities, preferreds, warrants, and non-common shares. Liquidity: average daily dollar volume $\geq$ \$100 million over the prior 30 trading days and average share volume $\geq$ 1 million. CURRENT FRIDAY SETUP: evaluate near the end of the regular Friday session. Candidate must be trading below Friday's regular-session open and preferably in the bottom 30% of Friday's high-low range, using close-location = (current price - day low)/(day high - day low). Full Pass current setup requires close-location $\leq$ 0.30; Watchlist may include $\leq$ 0.40 if the historical edge is unusually strong. Report Friday open-to-current return and close-location percentile. HISTORICAL TEST: use approximately the prior 2 years of trading data where available, emphasizing the most recent 1 year. For each historical Friday that meets the weak-Friday condition (Friday close < Friday open and Friday close-location $\leq$ 0.30), calculate: (1) Monday official open return = Monday open / Friday close - 1; (2) Monday official close return = Monday close / Friday close - 1; (3) whether Monday intraday high touched +1% and +2% versus Friday close; (4) Monday worst intraday drawdown versus Friday close. Skip holiday-shortened weekends where the next session is Tuesday unless separately labeled; do not mix Tuesday-after-holiday observations into the primary Monday statistics. Require at least 25 historical qualifying Friday observations for Full Pass. FULL PASS thresholds: Monday-open win rate $\geq$ 60%; Monday-close win rate $\geq$ 60%; average Monday-close return $>$ 0%; median Monday-close return $>$ 0%; probability of touching +1% on Monday $\geq$ 70%. RECENT PERSISTENCE: separately evaluate the most recent ~12 months and, when sample allows, the most recent ~6 months. Require recent Monday-close win rate $\geq$ 60% and recent average Monday-close return $>$ 0% for Full Pass. If long-term thresholds pass but recent persistence fails, classify as Watchlist. EARNINGS FILTER: exclude any stock with confirmed or projected earnings scheduled after Friday's close, over the weekend, or on Monday before the intended exit window. Use Massive earnings data when available; otherwise use reliable current sources, preferring company investor-relations information. If earnings timing cannot be verified, do not award Full Pass; classify as Watchlist with EARNINGS DATE/TIME UNVERIFIED. EVENT RISK: flag known material/binary events over the weekend or Monday morning. TAIL-RISK CONTROL: report worst Monday-close return and preferably 5th-percentile Monday-close return; penalize patterns dominated by a few large positive gaps or with severe downside tails. RANKING: rank by Monday-close win rate, Monday-open win rate, recent persistence, median and average Monday-close return, +1% Monday hit rate, sample size, liquidity, and downside-tail behavior. REPORT for each candidate: ticker/company, market cap, Friday current price/open/high/low, open-to-current %, close-location %, historical sample count, Monday-open win rate/average return, Monday-close win rate/average/median return, +1% and +2% Monday hit rates, worst and 5th-percentile Monday return, recent sample/win/average stats, next earnings date/time/status, event-risk notes, liquidity metrics, and concise reason for ranking. Clearly classify Full Pass / Watchlist / Reject. If no Full Pass candidates exist, show the best near-misses and exactly which rules they missed. This is a statistical screening tool, not a prediction or guarantee; weekend gaps and unexpected news can cause losses larger than historical averages.

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3-5 Day Swing Scanner (On Demand)

Run Dennis's 3-5 Day Swing Scanner using connected Massive market data as the primary source. Goal: rank liquid U.S.-listed common stocks with the strongest evidence of upside continuation over the next 3-5 trading days. Exclude ETFs, leveraged/inverse products, OTC securities, preferreds, warrants, units, and

non-common shares. Prefer price $\geq \$5$, 30-day average share volume ≥ 1 million, and 30-day average dollar volume $\geq \$100$ million. Calculate recent 1/5/10/20-day returns, 10-day and 20-day moving averages, 20-day high, 10-day low, RSI(14), close location, VWAP context when available, and relative volume. Prefer positive 5-day momentum, price above the 10-day and 20-day averages, a close within roughly 3% of the 20-day high or a clean breakout, strong close location, relative volume $\geq 1.2x$, and RSI roughly 50-68; penalize RSI >75 unless a fresh catalyst justifies the extension. Reconstruct comparable historical signals without look-ahead and measure forward 3-day/5-day win rates, average/median return, +1%/+2% touch rates, worst return, and preferably 5th-percentile return. Normally require $\geq 60\%$ 5-day positive rate with positive average and median for Full Pass, subject to a reasonable sample size and recent persistence. When options are available, analyze near-term call/put volume and open interest as confirmation; call/put volume $>1.5x$ is supportive and $>2.5x$ is strong when liquid and broad-based. Review recent Massive news/sentiment and reward fresh positive catalysts while identifying conflicting news. Compare candidates with SPY/QQQ and the relevant sector proxy where practical. Exclude Full Pass when earnings fall inside the intended holding window unless the user explicitly asks for an event trade. Flag major binary company events and macro events such as FOMC, CPI, or payrolls during the holding window. Rank using approximately: trend/momentum 25%, breakout/close quality 15%, volume/liquidity 15%, historical persistence/downtail 15%, options confirmation 10%, catalyst/news 10%, market/sector relative strength 5%, extension/gap/RSI risk 5%. Full Pass generally requires score ≥ 75 and no hard veto; 65-74 or material caveats are Watchlist; below 65 or hard-gate failures are Reject. For each top candidate report rank, ticker/company, score/grade, current/close price, technical and volume metrics, historical 3-5 day statistics, options confirmation, catalyst, event risks, preferred entry zone, breakout trigger, 3-5 day target zone, invalidation level, and no-chase warning where applicable. Do not automatically buy the opening print; for opening scans prefer roughly 15-30 minutes of confirmation. Recalculate after gaps $>3\%$, and normally do not chase gaps $>5\%$ without a fresh favorable risk/reward assessment. Return the top five Full Pass candidates plus a short Watchlist, and state clearly when no stock meets the criteria. This is a statistical screening tool, not a prediction or guarantee.

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90-Day / 15% Downside Safety Scanner (Paused)

Run Dennis's 90-Day / 15% Downside Safety Scanner for liquid U.S.-listed common stocks and conventional ETFs. Use connected Massive.com as the primary source. Exclude OTC, preferreds, warrants, units, closed-end funds, leveraged/inverse ETFs, volatility products, single-stock ETFs, synthetic high-leverage products, and ETNs. Require price \$5-\$1,000, 30-session average volume ≥ 1.5 million shares, average daily dollar volume $\geq \$100$ million, listed options, and reliable adjusted history. Use about 10 years of data when available, minimum five years. For every historical start date, test whether the lowest adjusted daily low during the next 63 sessions remained strictly above 85% of the start-date adjusted close; an exact -15% touch is a breach. Report full-history, recent-five-year, non-overlapping, Wilson-bound, VIX-regime, below-200-day-MA, crisis, worst-MAE, and tail statistics. Tier A requires full-history and recent-five-year survival $\geq 95\%$ plus a non-overlapping 95% Wilson lower bound $\geq 90\%$; Tier B requires both primary rates $\geq 90\%$; Tier C requires full-history $\geq 85\%$. Add current technical/risk metrics and an optional 30-120 DTE put-credit-spread overlay prioritizing 75-105 DTE, short strike $\leq 85\%$ of spot, net delta tiers, OI ≥ 500 per leg, credit $\geq \$0.25$, RoR $\geq 5\%$, and the seven-day earnings buffer. Verify live Robinhood pricing before any trade. After every completed run, email the complete report to Dennis Weng at dwengca@gmail.com with subject '90-Day Downside Safety Scanner - [run date in America/Los_Angeles]', including when no Tier A candidates exist.

Use at your own risk. Historical survival rates are not guarantees. Verify all results, data freshness, earnings/event timing, liquidity, and live broker pricing before acting.